B.Comp. Dissertation

Research in blockchain and cryptocurrency – DePLOB (Decentralised Private Limit Order Book)

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1. Introduction

1.1 Problem Motivation

Blockchain technology provides the foundational infrastructure for decentralised finance ("DeFi") by maintaining a distributed, immutable ledger of transactions that is not controlled by any single intermediary - commonly referred to as decentralisation in the blockchain space. In this paradigm, users interact with smart contracts, autonomous programs deployed on the blockchain that execute predefined financial functions (such as lending, trading or settlement) once specified conditions are met (Benetti & Piazza, 2024)[1]. Each transaction is broadcast to the network, enters a pool of pending operations, is validated (via consensus mechanisms such as proof-of-work or proof-of-stake) and, upon inclusion in a block, becomes part of the chain and are then visible and verifiable by any participant[2].

Importantly, because all transaction data such as wallet addresses, amounts, contract calls and state changes are recorded on the public ledger, the system is pseudonymous yet transparent: as users on the blockchain would not be tied to their real life identity but their wallet address. It is to note that a user can control multiple wallet addresses at once due to the same decentralised nature of blockchain.

In decentralised finance, trading mechanisms broadly fall into two paradigms. One is the automated market-maker (AMM) or constant-function market-maker (CFMM) model, in which liquidity providers deposit assets into smart-contract pools and trading occurs by interacting with those pools via a deterministic pricing rule (e.g., $x \times y = k$ in the constant-product model).[3] The alternate paradigm is a central limit order book (CLOB), which organises outstanding limit buy and sell orders and matches them according to price‑time priority. While CFMMs have

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[1] https://www.esma.europa.eu/sites/default/files/2024-08/ESMA50-43599798-9913_Working_Paper_Decentralised_Finance_-_A_Categorisation_of_smart_contracts.pdf

[2] https://www.sciencedirect.com/science/article/abs/pii/S2352673419300824

[3] https://arxiv.org/pdf/2105.02782

dominated many DeFi platforms due to their simplicity and composability, CLOBs offer richer trader functionality (limit orders, fine‑grained price control, deep order depth) and more efficient capital usage.

However, due to the transparent nature of blockchain, every transaction, whether a liquidity pool swap, order placement, cancellation or match - results in a publicly visible state transition on the ledger and incurs a gas fee paid by the user. This means the entire order book state, its updates, the identities of address‑actors (to some extent), and the amounts involved are visible to observers unless additional privacy measures are applied. For example, in an on-chain CLOB, order placements and cancellations reside in publicly readable contract storage or events, enabling visibility into market depth, trader behaviour, and potential front-running by adversarial miners or sequencers.

1.2 Industry Gap: Privacy

1.2.1 Importance of Privacy

Privacy in decentralized exchanges is a critical protection for traders, especially those executing large orders or leveraged positions. The inherently public nature of blockchain transactions enables adversaries to observe pending transactions, infer trader intentions or liquidation thresholds, and exploit that information via front-running, sandwich attacks or coordinated liquidation hunts. In fact, in their seminal work on DEX adversarial behaviour, Daian et al. (2019) show that bots routinely monitor broadcast transactions and compete via priority-gas-auctions to insert themselves ahead of honest traders, thereby extracting value (MEV) and degrading market fairness[4].

Furthermore, recent commentary from Changpeng Zhao ("CZ") (2025) underscores the demand for privacy-focused trading infrastructure: "I have always been puzzled with the fact that everyone can see your orders in real-time on a DEX… If others can see your liquidation point, they could try to push the market to liquidate you." (CZ, 2025)[5]

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[4] https://arxiv.org/abs/1904.05234
[5] https://www.cryptotimes.io/2025/06/02/binance-cz-nows-time-for-a-privacy-focused-perpetual-dex/

A concrete example is the dramatic liquidation event on Hyperliquid, where transparent order flows and visible liquidation levels reportedly facilitated large-scale liquidations of short positions triggered by sudden price moves (Hyperliquid incident, 2025)[6] These realities reinforce the research finding that "transaction-inspection" on public ledgers significantly increases the risk of front-running and market manipulation in DEXes (Heimbach et al., 2023)[7] Therefore, enabling confidentiality of order amounts, identities and order-book state is not just an optional enhancement but a prerequisite for providing fair, efficient and secure trading in on-chain order-book systems.

Centralised exchanges, while often offering operational privacy in the sense that users' individual trades are handled off-chain and not directly exposed to the public ledger, remain fundamentally **not decentralised**. In a typical custody-based model, users deposit their assets into the exchange's wallets and trades are matched within the exchange's internal order-book rather than directly on‑chain (Zexe et al., 2018)[8]. This design grants the exchange full control over assets, order execution, withdrawal permissions and account data transforming the user into a trusting counterparty rather than an autonomous participant. Although such platforms may shield user trading details from on-chain visibility, they concentrate trust and risk in a single entity: the exchange operator. As a result, the arrangement trades away decentralisation, self-custody and censorship-resistance in exchange for throughput and convenience which defeats the purpose of the DeFi movement.

1.2.2 Dark Pools

A dark pool is an alternative trading venue, typically an off-exchange, non-lit order-match system that allows participants, often institutional investors, to execute large trades without publicly revealing their order size, price or identity until after execution.[9] These venues emerged to enable block trades with reduced market impact and reduced information leakage relative to daytime public exchanges. Because pre-trade order book data is not visible to the broader market, dark pools operate with limited transparency and are subject to scrutiny regarding their effects on price discovery and fairness.

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[6] https://www.cryptomarketwatch.pro/2025/09/28/inside-hyperliquids-biggest-eth-liquidation-defis-stress-test/
[7] https://arxiv.org/abs/2306.05756
[8] https://eprint.iacr.org/2018/962.pdf
[9] https://www.investopedia.com/articles/markets/050614/introduction-dark-pools.asp?utm_source=chatgpt.com

1.3 Objective of Research

In this regard, the transparent nature of blockchains poses a fundamental challenge for privacy‑sensitive trading systems, especially for actors requiring hidden position size, counterparty anonymity, or confidential order books.

Therefore there is a strong demand for an on-chain decentralised private limit order book that allows users to trade privately. This inherently means that the private limit order book exchange must have a provably correct order creation and matching system.

2. Literature Review

2.1 Tornado Cash

Tornado Cash (henceforth Tornado) was launched in December 2019 and is one of the earliest privacy protocol and most widely used non-custodial mixing protocols on the Ethereum network (Pertsev et al., 2019)[10]. Its design centres on a smart contract that accepts fixed-denomination deposits (e.g., 10 ETH) and, at some later point, allows a withdrawal to a different address for the same denomination, but unlinked from the depositing address. The cryptographic engine uses commitments to a secret note (generated off‑chain) and zero‑knowledge SNARK proofs to assert knowledge of a valid deposit leaf in a Merkle commitment tree, along with a unique nullifier, without revealing which leaf is being spent. (Pertsev et al., 2019) The protocol ensures that no link is revealed between deposit and withdrawal, so long as the anonymity set is sufficiently large. In that sense, Tornado offers strong value to users seeking confidentiality of transaction flows in a public-ledger environment.

From the perspective of privacy primitives, Tornado's architecture is highly relevant to our design agenda. First, it demonstrates a working integration of zero‑knowledge cryptography (e.g., Pedersen hashes, MiMC sponge‑hashes, zk-SNARK proofs) into an EVM-deployed mixer smart contract (Pertsev et al., 2019). Second, it shows how a shielded pool used for anonymisation (of funds transfers) can be constructed in a non-custodial, on-chain fashion as

[10] https://berkeley-defi.github.io/assets/material/Tornado%20Cash%20Whitepaper.pdf

users retain control of their assets (no intermediate service custody) and the mixer smart contract simply enforces the cryptographic logic. Third, the model of fixed-denomination coins, Merkle-tree commitments and nullifiers is a canonical building block which is already adopted in many mixing/shielded-pool systems (and can be adapted for private order-book scenarios). For our work, the idea of mapping each order (or order-note) into a shielded commitment, then later revealing (or matching) it with privacy, builds on exactly this model.

However, Tornado has a few limitations in its model that motivate the design choices we make in our system (DePLOB). One of the most discussed issues is anonymity-set shrinkage and linkability: while Tornado assumes that each withdrawal is effectively unlinkable among all deposits in the pool, multiple empirical studies have shown that heuristics (e.g., address-reuse, "first‑in first‑out" temporal correlation) can significantly undermine the anonymity guarantees. For example, Tang et al. (2021)[11] apply clustering heuristics to Tornado data and demonstrate that a non-trivial fraction of addresses can be linked from deposit to withdrawal patterns, reducing effective anonymity. This implies that even if the cryptographic primitives are sound, real‑world user behaviour (such as re-using addresses, fixed denominations, or predictable timings) can degrade privacy. In designing DePLOB, we therefore pay explicit attention not just to cryptographic confidentiality of orders, but also to routing/mixing logic, multi‑token pooling and order-book hiding to avoid similar leakages.

Another limitation of Tornado is its application domain: it is a transfer mixer. Funds go in, then later (to a different address) they come out. There is logic to swap tokens built in. While Tornado satisfies the requirement of hiding origin and destination of funds, it does not address the functionalities of order-book exchanges. In our literature survey (Section 2), we see that while Tornado addresses a key primitive (shielded pool), it leaves a large gap in the domain of on-chain private order-book exchange. That gap is precisely where DePLOB is positioned.

In summary, Tornado serves as a foundational reference for privacy-preserving funds-transfer on-chain providing key cryptographic and systems building blocks. Yet its focus remains restricted to transfers (not orders), its anonymity guarantees are weakened by real-world user patterns, and it does not directly support order-book semantics. The design of DePLOB builds on

[11] https://link.springer.com/content/pdf/10.1007/978-981-16-9229-1_3.pdf

Tornado's strengths (shielded pool, nullifier/commitment model, non-custodial smart contract) but extends the vision to continuous-order-book trading with multi-token pooling, routing and sub-block latency, thereby filling a clear research and system-design gap.

2.2 BlockAuction

BlockAuction proposes a privacy-preserving and verifiable periodic double auction mechanism that combines encrypted order submission, matching and on-chain settlement (Constantinides & Cartlidge, 2021)[12]. In this framework, participants submit sealed/encrypted bids or asks, which are then processed in bulk at periodic clearing intervals; the results (winner selections, clearing prices) are posted on-chain, enabling public verifiability of the outcome without necessarily revealing individual bid values. The protocol emphasises cryptographic confidentiality of bids combined with on-chain auditability of the matching process. Its design therefore situates itself between classic sealed-bid auctions and continuous trading platforms with order books.

BlockAuction offers several architectural insights that are directly relevant to the DePLOB design agenda. First, it demonstrates that encryption of orders (or bids) combined with periodic matching can be executed in a blockchain-smart-contract context, showing how confidentiality and on-chain settlement can coexist. Second, the use of bulk matching rather than continuous streaming allows the system to batch operations, reducing per‑transaction overheads and potentially improving performance (a useful referent when considering 1-block latency targets). Third, by addressing verifiability (i.e., publishing sufficient outputs on-chain for anyone to audit that the matching rule was applied correctly) it aligns with our requirement of "on-chain verifiability" for DePLOB's matching and routing logic.

Despite these strengths, BlockAuction leaves several important gaps when viewed from the perspective of a high-performance, fully decentralised, private continuous limit-order-book (CLOB) on-chain system. Firstly, its periodic clearing model implies that orders must wait until the next batch rather than being consumed on a per-block basis introduces unwanted latency. Secondly, while confidentiality of bids is maintained, the design does not evidently support full order-book depth hiding (e.g., of limit orders, visible book state, incremental cancellations) or

[12] https://research-information.bris.ac.uk/ws/portalfiles/portal/300674199/21.11.01.BlockAuction.AAM.pdf

continuous route/multi-token trading across pairs. Lastly and most importantly, the trust assumptions often still include a central or semi-trusted coordinator or matching authority. By contrast, DePLOB aims to replace classical matching authority with a decentralised TEE-based decryption oracle and a singleton smart contract architecture supporting multiple token pairs within a single shielding pool.

BlockAuction shows how encrypted orders + on-chain settlement can be architected in a blockchain context. However, changes could be made to avoid the central matching authority which can be compromised. In doing so, we bridge the gap between sealed‑bid/batch auction systems and full-featured private CLOBs suitable for a decentralised finance setting.

2.3 Renegade

Renegade is a non-custodial, on-chain "dark-pool" exchange protocol that became live on Arbitrum in September 2024. The key innovation is that orders and trades remain private from the order's initiation through to settlement: no external observer, including block proposers or validators, should be able to infer trading details, participant identities or order-book state. It uses a combination of secure multi-party computation (MPC) for matching, and zero-knowledge proof (ZKP) based settlement, thereby enabling peer-to-peer anonymous crossing of orders (rather than a publicly-visible live order-book). The protocol emphasizes midpoint pricing (e.g., clearing trades at the midpoint of another major exchange's bid-ask) to avoid slippage and price impact for large trades.

Renegade's architecture is highly relevant to a privacy-preserving on-chain limit order book (CLOB) system such as DePLOB for several reasons. First, it demonstrates that full trade-lifecycle privacy (pre-trade order details, on-trade matching, and post-trade settlement) is feasible in a permissionless blockchain context using cryptography rather than trusting a central intermediary. Such an end-to-end private trading model addresses many of the information-leakage issues inherent in transparent order-books on chain. Second, the use of a dark-pool style crossing network (rather than typical AMM or public CLOB) offers a feasible path for large-size trades without exposing order flow or triggering front-running, MEV attacks

or quote-fading effects. Third, the fact that Renegade is live on a blockchain environment (Arbitrum) means its design and implementation provide practical evidence of how private trading venues might be integrated into DeFi ecosystems even if the model differs from a full continuous order-book. Therefore, it serves as a useful benchmark for our DePLOB project's goals of hiding order book contents, trader identities, and trade amounts.

Despite its strengths, Renegade also leaves open gaps especially when contrasted with the goals of a fully featured, decentralized private order-book like DePLOB. One limitation is that Renegade is fundamentally a dark-pool / crossing-network model rather than a continuous limit-order book with visible depth, price-time priority, cancellations and routing across multiple token pairs. While dark pools remove visibility of the full order book, this also means less access for broader price-discovery, which may affect liquidity and market fairness. Second, the trust and performance assumptions may differ: although Renegade emphasises MPC and ZKP (avoiding hardware TEEs), the matching is still dependent on some coordinated infrastructure (e.g., a peer-to-peer gossip network for order matching) which may impose latency or scalability trade-offs. Lastly, although Renegade promises "zero MEV" and "midpoint pricing", the fact that matching is contingent on sufficient counter-flow implies that during skewed market conditions, execution might experience latency or risk undesirable time to fill.

In effect, while Renegade addresses many privacy challenges, it does not yet fulfil the full functional envelope of a fully private CLOB with multi‑token routing inside one shielded pool which is precisely the gap our design aims to achieve.

2.4 Lighter

Lighter is an order-book based decentralized exchange platform built as an application-specific zk-rollup on Ethereum (or an L2 variant) and designed to deliver high-throughput, verifiable matching and settlement via zero-knowledge proofs (Elliot Technologies, Inc., 2025)[13]. The protocol supports a central limit order book (CLOB) architecture with price-time priority matching of orders, but wraps the matching engine within a zk-SNARK proof system so that the correctness of the matching, liquidations, and state transitions can be verified on-chain. In

[13] https://assets.lighter.xyz/whitepaper.pdf

practice, Lighter emphasises features such as zero trading fees for retail users, sub-millisecond latency matching and a non-custodial design where users retain control of their funds.

Lighter illustrates several design choices that are highly informative for a privacy-preserving on-chain order-book system like proposed “DePLOB”. First, it demonstrates that a full CLOB architecture, orders, matching, cancellations, and liquidations, can be implemented in a scalable, high-throughput environment on chain (or via rollup) while preserving auditability via zk-proofs. Second, its verifiable matching logic addresses one of the key trust issues in decentralised exchange infrastructure: how to ensure that the matching engine honours price-time priority and resists front-running and unfair ordering without relying solely on external trust in sequencers. Third, the non-custodial design combined with advanced cryptography helps approximate the self-custody and decentralisation goals that are core to DeFi and to DePLOB’s ambition of hiding trader identity/amounts while remaining transparent at the settlement layer.

Despite these advances, Lighter also exhibits limitations when viewed against the specific requirements of a fully private and decentralised CLOB with multi-token routing and 1-block latency (as in DePLOB). One limitation is that while Lighter offers verifiable matching, it does not emphasise privacy of order amounts, identities or order-book depth as part of its core design as its focus is on fairness and low latency rather than confidentiality. Second, its architecture (a dedicated zk-rollup matching engine) may still rely on sequencer/centralised components or batching processes which could introduce latency or trust assumptions; for DePLOB’s goal of one-block latency and minimal trust surface (via TEE-based oracle), this remains a gap. Third, while Lighter supports a CLOB for one or several pairs, it does not explicitly present a “shielded pool” abstraction with multi-token routing inside a single contract that hides the full book state and allows private matching across many token pairs within the same pool. Hence, the private trading and routing novelty of DePLOB remains unaddressed by Lighter. Additionally, although Lighter demonstrates strong engineering claims (e.g., thousands of orders per second, sub-5ms latency) under favourable conditions, real-world stress events like cascading liquidations in October 2025 have exposed vulnerabilities like outages which highlight the operational risk of roll-up/sequencer-centred architectures. (Njri, 2025)[14]

[14] https://ambcrypto.com/lighter-dex-to-repay-users-in-stablecoins-after-major-outage-details/

From the analysis of Lighter, several design lessons emerge for DePLOB. The use of zk-proofs for matching and settlement suggests that on-chain verifiability of ordering logic is feasible and desirable. However, to achieve DePLOB's goals of strong privacy, we must integrate additional architectural elements beyond Lighter's scope. Specifically, combining Lighter's high-performance CLOB architecture with shielded-pool primitives (like those in privacy mixers) and private order encryption/decryption workflows (as in MPC/TEE designs) is required. Moreover, operational resilience (to stress events), decentralised oracle/trust assumptions, and behavioural soundness (preventing linkability via timing/amount correlations) when shifting from verifiable fair matching to privacy-enhanced matching must also be considered.

2.5 Rialto (IBM)

Rialto is a fully decentralised, privacy-preserving marketplace protocol designed for exchange markets on a blockchain (Govindarajan et al., 2021)[15]. It supports order-matching, on-chain settlement and market price discovery, while aiming to preserve confidentiality of order rates and account balances, and unlinkability between traders and their trade orders. The authors formalise security notions such as confidentiality of order parameters and unlinkability of trader identities, then propose an architecture based on MPC (secure multi‑party computation) among brokers plus cryptographic commitments and mixing techniques to hide links between orders and participants. They evaluate the system and conclude it can scale to a certain class of goods and financial instruments.

First, Rialto shows that order books and matching logic need not sacrifice privacy entirely: by hiding order rates and balances, the system protects key trader metadata while retaining a marketplace function. Second, it emphasises on-chain settlement but with privacy protections layered on top offering a blueprint for how one might integrate private matching with blockchain settlement. Third, by defining formal security goals (confidentiality, unlinkability, fairness) and providing a realistic evaluation, Rialto contributes to the methodological foundation of privacy‑preserving exchange systems.

[15] https://arxiv.org/pdf/2111.15259

Despite its merits, Rialto has several limitations. Firstly, the use of MPC among brokers introduces additional coordination overhead and trust assumptions namely that a threshold of brokers behaves honestly which may impair latency or decentralisation in a fully permissionless setting similar to BlockAuction from Constantinides et al (2022). Finally, the evaluation, while promising, does not deeply explore on‑chain proof costs, composability with EVM contexts or the shielding of full order-book state (i.e., the depth of the book remains visible) which questions the practicality of Rialto.

2.6 COMMON

The COMMON protocol proposes a privacy-preserving on-chain order‑book DEX architecture built around a shielded pool and decryption oracle model (Garreta et al., 2023)[16]. Users deposit funds into a shielded anonymity pool, submit encrypted orders, and matching is performed within that pool without immediate public disclosure of order details. Subsequently, a decryption oracle (via threshold decryption/MPC) reveals just enough information to settle matched trades on-chain while preserving confidentiality of participant identities and order amounts. The design separates three phases: “anonymity pool”, “execution engine”, and “decryption oracle”. (Garreta et al., 2023) It thereby aims to combine continuous order‑book semantics with strong privacy, on-chain verifiability and efficient settlement.

COMMON’s architecture offers several very compelling features. First, by embedding a shielded pool at the core, it shows how order placement, matching and settlement can be decoupled from the publicly readable state of a conventional on-chain order book. This supports hiding user identity, order size and standing order-book depth. Second, the use of encrypted orders and a decryption oracle illustrates that smart-contract based matching can occur while preserving confidentiality bridging the gap between classic dark-pool models and on-chain order-books. Third, the protocol maintains on-chain verifiability: even though orders are encrypted, the matched outcome is recorded on-chain and audit trails are retained, supporting transparency of settlement without exposing sensitive order details. These features align closely with our objectives of on-chain verifiability and privacy.

[16] https://eprint.iacr.org/2023/1868.pdf

Despite its merits, COMMON exhibits limitations which motivate the design extensions in DePLOB. One key limitation is the over-reliance on an MPC‑based decryption oracle, which introduces trust and coordination assumptions (a threshold of parties must behave honestly) and may incur latency or complex operational overhead. DePLOB aims to replace this with a decentralised TEE-based oracle, reducing the trust surface and improving latency. Second, while COMMON provides a shielded pool, the whitepaper emphasises per‑token discrete pools and less focus on a unified multi-token routing architecture inside a singleton smart contract; DePLOB's novelty is precisely that integration of multi-token private trading and routing within a single shielded pool and contract. Third, although the whitepaper targets privacy, the practical performance and implementation remains untested. Lastly, the user-behaviour dimension (e.g., linkability via timing, amount correlations, pool size/ anonymity set) is not deeply addressed in COMMON.

From the above analysis, COMMON clearly provides an important foundation: a shielded pool model, encrypted orders with a decryption oracle, and on-chain settlement with verifiability. However, to fill the identified gap, DePLOB will build on COMMON by (1) replacing MPC with a decentralised TEE-based oracle to improve trust and latency, (2) designing a singleton smart contract architecture that supports multi‑token trading and routing inside a single shielded pool, (3) allowing on-chain verifiability of matching and routing, and (4) enhancing mitigation of linkability and behavioural side‑channels (timing/amount correlations) to better preserve privacy in a realistic operational environment.

3. DePLOB (Decentralised Private Limit Order Book)

3.1 Progress Made So Far

The progress made so far has focused on defining and designing the core functionality required for a decentralised private limit order book (DePLOB). The system's overarching goal is to reconcile the expressivity of a traditional central-limit-order-book (CLOB) with the privacy guarantees of modern shielded-pool protocols. At its foundation, DePLOB is implemented as a singleton smart contract managing a unified shielded pool of assets across multiple token pairs. This contract acts as both the custody layer and the state machine for deposits, withdrawals, and private order lifecycle management. By maintaining a single pool rather than pair-segmented liquidity, DePLOB enables private multi-token routing while simplifying proof verification and gas accounting on the EVM.

The first major functionality achieved is the secure deposit and withdrawal mechanism within the shielded pool. Each user interaction begins with a deposit transaction that commits encrypted asset information into the pool via a zero-knowledge proof. Users generate cryptographic commitments representing ownership of assets, along with nullifiers that prevent double spending. Withdrawals are executed through corresponding proofs demonstrating possession of a valid, unspent commitment without revealing which deposit it originated from. This ensures that on-chain observers cannot correlate deposits and withdrawals, preserving user anonymity and asset confidentiality. This design inherits structural inspiration from protocols such as Tornado Cash but extends it to support arbitrary ERC-20 assets within a unified, multi-token context.

The second functionality under active prototyping is the private order management subsystem. Users can submit, modify, and cancel limit orders directly from within the shielded pool. Each order is encrypted using the system's public parameters, concealing critical information such as price, amount, and token pair. Orders are stored as encrypted state commitments in the contract's Merkle tree, maintaining verifiable but hidden order-book state. To cancel or update an order, users present a zero-knowledge proof of ownership over the corresponding commitment,

ensuring that no party, including validators or sequencers, can infer the nature or timing of their trading intent. The system's architecture supports order batching and ephemeral keys for additional unlinkability across user actions.

Finally, DePLOB will incorporate a private and verifiable matching mechanism, designed to perform the functional equivalent of price–time priority matching within a privacy-preserving context. The matching engine operates off-chain under encrypted state and interacts with a decentralised TEE-based decryption oracle, which replaces the MPC-based oracle used in COMMON. This oracle collaboratively decrypts matched orders within a trusted execution environment and posts verifiable proofs of correctness on-chain. This design ensures that matched trades are both private as only matched orders are selectively decrypted, and verifiable, as every decryption and settlement event is publicly auditable through ZK attestations and TEE attestation proofs. The combination of shielded state, private order lifecycle, and verifiable matching enables DePLOB to provide one-block latency while maintaining full user anonymity and market integrity.

In summary, the system now comprises three integrated components

- a shielded pool with private deposits and withdrawals,
- a private encrypted order-book interface for submission and cancellation
- a verifiable matching system employing decentralised TEEs

Together, these form the minimal viable architecture of DePLOB: a privacy-preserving, on-chain, verifiable, multi-token limit-order-book exchange that achieves continuous trading semantics without sacrificing decentralisation or transparency.

In defining the security perimeter for DePLOB, we consider a semi-honest to malicious adversarial model, where participants including traders, validators, and matching oracles may attempt to infer private information, manipulate order execution, or disrupt protocol fairness to extract value from the system. The goal of the system is to maintain confidentiality, integrity, and verifiability even in the presence of such adversaries.

The primary adversary class comprises honest-but-curious participants entities that follow the protocol but attempt to analyse on-chain data, transaction timings, or ciphertext patterns to infer

user identities, order sizes, or trading intent. Because Ethereum's transaction layer is transparent, even metadata such as gas usage, nonce ordering, or event emissions can reveal information about hidden actions. DePLOB's shielded pool architecture mitigates this by replacing direct token transfers with cryptographic commitments and nullifiers. Each deposit, order, and withdrawal is represented as a ZK commitment, ensuring that transaction values and ownership are not linkable. However, side-channel leakages such as transaction timing, gas consumption variability, and block-level ordering patterns remain a residual threat, especially to smaller anonymity sets.

The second adversary class includes malicious network actors such as miners, sequencers, or validators seeking to extract maximal extractable value (MEV) through front-running or order reordering. In transparent DEX architectures, such adversaries can observe pending transactions in the mempool and reorder them for profit. In DePLOB, encrypted order submission and a single-pool matching model prevent adversaries from observing specific trade parameters, thereby eliminating the feasibility of classical front-running and sandwich attacks. Nevertheless, if an attacker could correlate encrypted transaction timing or frequency with external market data, partial inference attacks could still occur. To mitigate this, DePLOB adopts uniform transaction sizes, ephemeral keys, and batch-based matching to reduce correlational linkability between user activity and market events.

A third adversary class targets the decryption oracle and the trusted execution environment (TEE) infrastructure. Since DePLOB replaces the MPC oracle from COMMON with a decentralised TEE oracle, adversaries could attempt to compromise TEE enclaves, falsify attestation proofs, or collude to reconstruct decrypted order information. The system therefore assumes that at least a threshold of TEE nodes remain uncompromised and that attestation proofs are publicly verifiable on-chain. Even if a subset of TEEs behaves maliciously, their activity should not compromise the confidentiality of unexecuted orders. Additional protections include periodic key rotation, redundancy among TEE validators, and auditing of attestation proofs to detect anomalies.

Finally, user behaviour represents an often-overlooked but critical threat vector. Even with perfect cryptographic privacy, careless operational patterns such as address reuse, deterministic

order timing, or correlated deposit and withdrawal sizes can erode effective anonymity. Prior analyses of Tornado Cash and other mixers (Tang et al., 2021) demonstrate that real-world anonymity often falls short of theoretical bounds due to behavioural heuristics. DePLOB addresses this by enforcing best practices at the protocol level, including non-reusable nullifiers, enforced cooldown intervals, and randomised batching for order decryption. However, ultimate privacy depends on maintaining a large and diverse active user set as anonymity is a collective property.